

Audio-Based Emotion Classification Using Mel Spectrograms

Project Summary, Results and Potential Improvements

1. Project Overview

This project develops an audio-based emotion classification pipeline that converts speech recordings into Mel-spectrogram-based representations and uses machine-learning models to classify the emotional content of the recordings. The executed notebook is titled “Clean Mel Emotion Pipeline” and runs the `clean_mel_emotion_pipeline` pipeline.

2. Dataset and Processing

- 1,440 audio files were processed.
- 1,440 feature matrices were loaded/generated.
- 1,440 Mel-spectrogram images were produced.
- 1,440 band-feature files were produced.
- The resulting feature matrix contained 1,440 samples and 718 features.

The notebook demonstrates a consistent conversion from the source audio into visual Mel-spectrogram representations and numerical features. The exact low-level settings (for example, sample rate, FFT size, hop length and number of Mel bands) are not exposed in the notebook itself, so they are not specified here.

3. Machine-Learning Pipeline

The executed pipeline evaluates several conventional machine-learning classifiers using the extracted audio features. Validation accuracy is used to compare candidate models, followed by evaluation on a held-out test set.

Model	Validation Accuracy	Test Accuracy
RBF SVC (C=10)	0.7870 (78.70%)	0.7454 (74.54%)
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Logistic Regression	0.7222 (72.22%)	0.6620 (66.20%)
Linear SVC	0.7037 (70.37%)	0.6389 (63.89%)
Histogram Gradient Boosting	0.6991 (69.91%)	0.6620 (66.20%)
Extra Trees	0.6944 (69.44%)	0.6667 (66.67%)
Random Forest	0.6667 (66.67%)	0.6343 (63.43%)
KNN	0.5833 (58.33%)	0.6111 (61.11%)

The notebook reports the selected model as “best_single”, with a test accuracy of 0.74537 (approximately 74.54%). The strongest candidate results shown in the notebook are the two RBF SVC configurations, both with 78.70% validation accuracy and 74.54% test accuracy.

4. Mel-Spectrogram Representation

A Mel spectrogram represents the changing frequency content of an audio signal over time using a frequency scale designed around the way humans perceive pitch. In this project, the Mel-spectrogram images provide a visual representation of speech characteristics that can vary with emotional expression, while the extracted numerical features provide the machine-learning models with a fixed-size input.

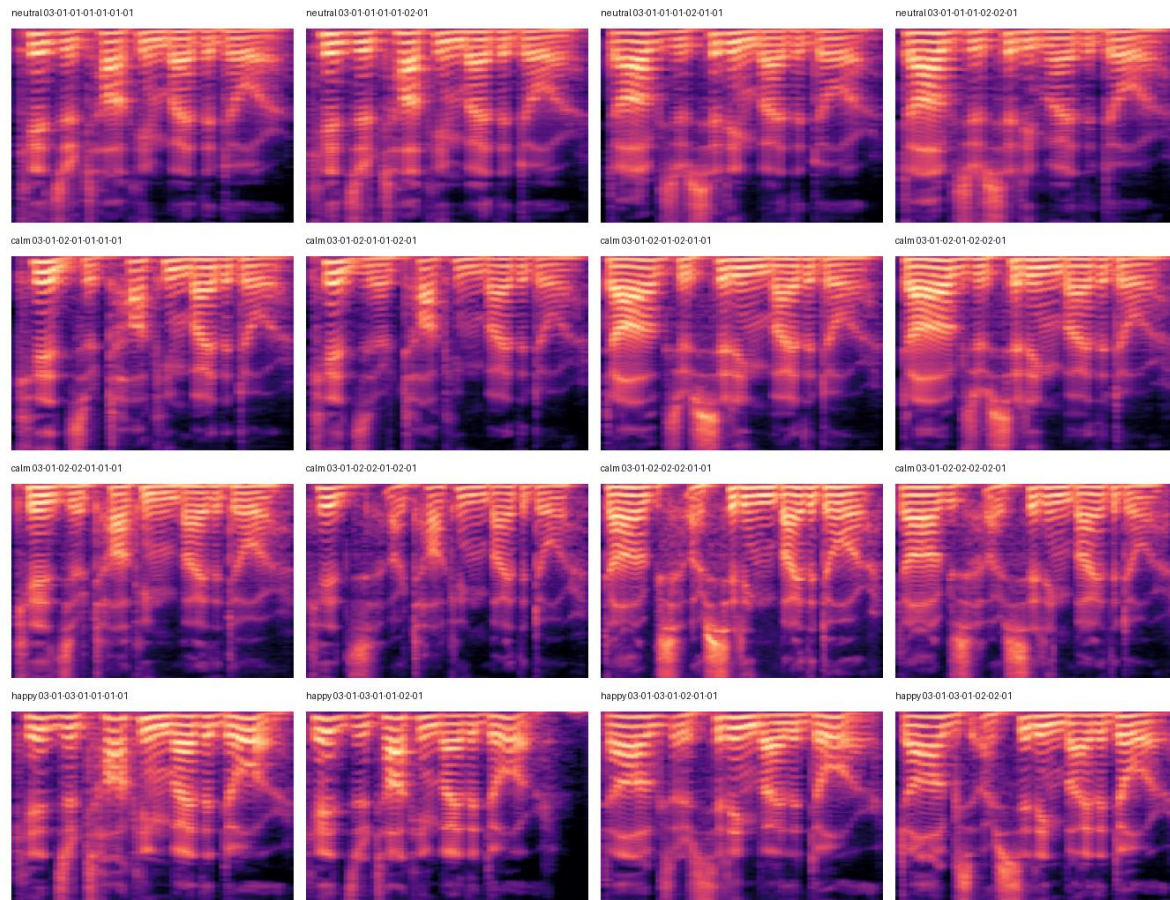


Figure 1. Mel-spectrogram examples produced by the executed pipeline.

5. Evaluation

The notebook produces both a model-metrics report and a confusion matrix. The test accuracy of about 74.54% indicates that the current feature-and-classifier pipeline can identify emotion classes with substantial predictive performance, while the difference between validation and test performance also shows that evaluation should remain a key part of further development.

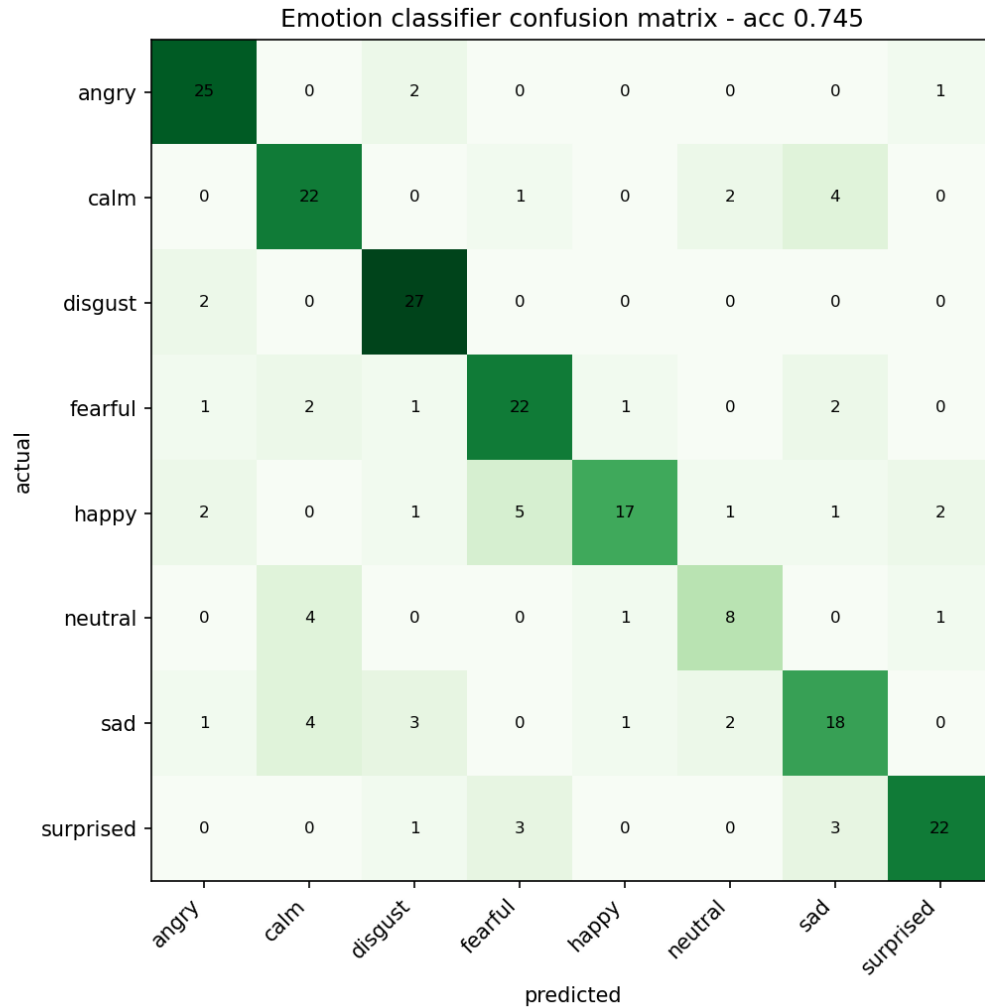


Figure 2. Confusion matrix generated by the executed pipeline.

6. Current Strengths

- End-to-end reproducible pipeline from audio files to extracted features and model evaluation.
- Complete processing of 1,440 audio recordings.
- Mel-spectrogram representations provide a meaningful time-frequency representation of speech.
- A 718-dimensional feature matrix is used for conventional machine-learning experiments.
- Multiple classifiers are compared rather than relying on a single algorithm.
- The pipeline generates quantitative metrics and a confusion matrix for evaluation.

7. Potential Improvements

Data augmentation: Add controlled audio augmentation such as time shifting, small pitch changes, background-noise injection, time masking and frequency masking. Augmentation should be applied only to the training portion to avoid leakage.

CNN-based Mel-spectrogram learning: Instead of relying only on engineered numerical features, feed Mel-spectrogram images directly into a 2D CNN so the model can learn local time-frequency patterns associated with different emotions.

Transfer learning: Evaluate pretrained audio models or spectrogram-based image backbones and fine-tune them on the emotion dataset. This can be useful when the available labelled dataset is relatively small.

Temporal modelling: Compare CNN-RNN, CNN-LSTM/GRU, or transformer-based approaches so that changes in emotion-related speech features across time are modelled explicitly.

Feature ablation: Measure the contribution of the current feature groups individually and in combinations. This can show which parts of the 718-feature representation actually improve classification.

Hyperparameter optimisation: Systematically tune SVC parameters such as C and gamma, alongside feature scaling and other model settings, using cross-validation rather than relying on a small manually selected set of configurations.

Class-wise evaluation: Report precision, recall, F1-score and support for every emotion class, in addition to overall accuracy. This is especially important when some emotions are harder to distinguish than others.

Speaker-independent validation: Ensure that recordings from the same speaker do not appear across training and test partitions. A speaker-independent split provides a stronger estimate of how the model may generalise to unseen speakers.

Robustness testing: Test performance under realistic conditions such as background noise, different recording volumes, microphone differences and shorter or longer utterances.

Reproducibility and experiment tracking: Record preprocessing parameters, random seeds, model hyperparameters, feature versions and dataset splits for each experiment so improvements can be compared reliably.

8. Suggested Future Pipeline

A stronger next-stage architecture could be: Audio Input → Quality Check → Train/Validation/Test Speaker Split → Mel-Spectrogram Generation → Training-Only Augmentation → Normalisation → CNN or CNN+Temporal Model → Class Prediction → Confusion Matrix + Precision/Recall/F1 → Error Analysis.

9. Conclusion

The executed work establishes a functional audio-emotion classification baseline using Mel-spectrogram representations and conventional machine-learning models. Across 1,440 audio files, the pipeline generated a 718-feature representation and achieved approximately 74.54% test accuracy with the reported selected single-model configuration. The clearest next steps are stronger data augmentation, speaker-independent evaluation, class-wise metrics, systematic hyperparameter tuning, and experiments with CNN or pretrained audio models that can learn directly from time-frequency representations.

Source note: Project-specific figures and model results in this summary are taken from the supplied executed notebook. The improvement recommendations are proposed next steps and are not claims that those methods were already implemented in the notebook.